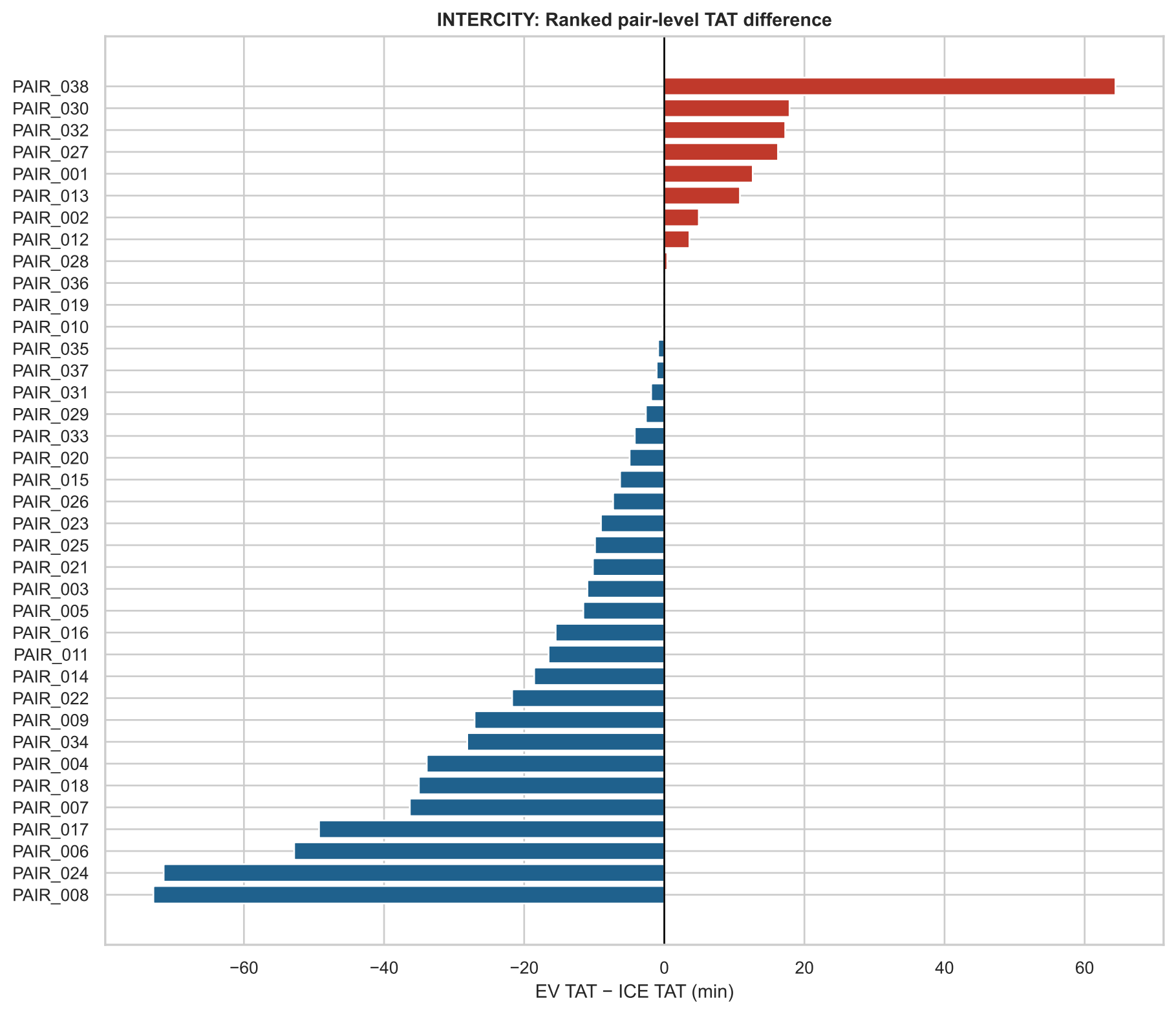
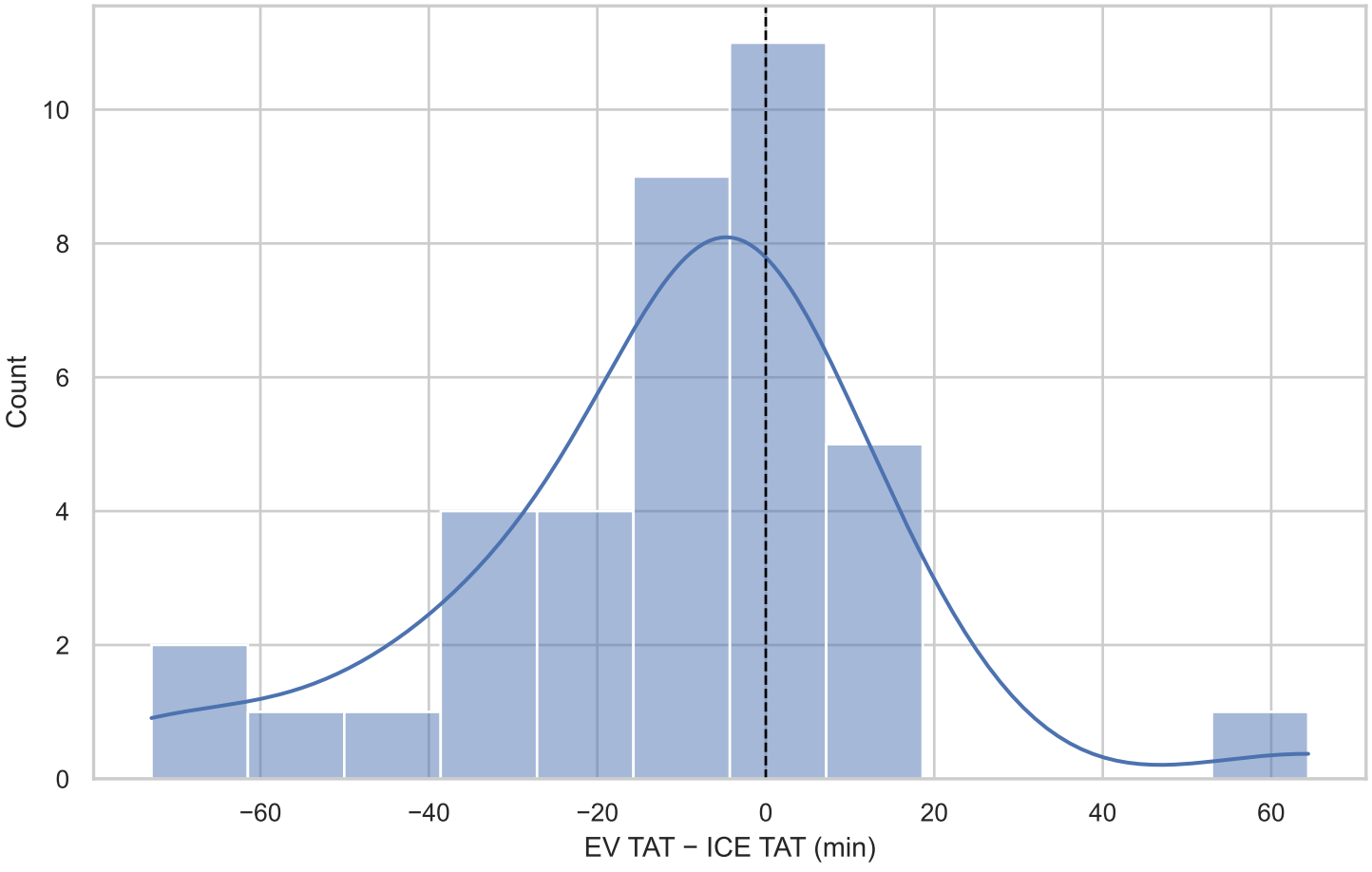
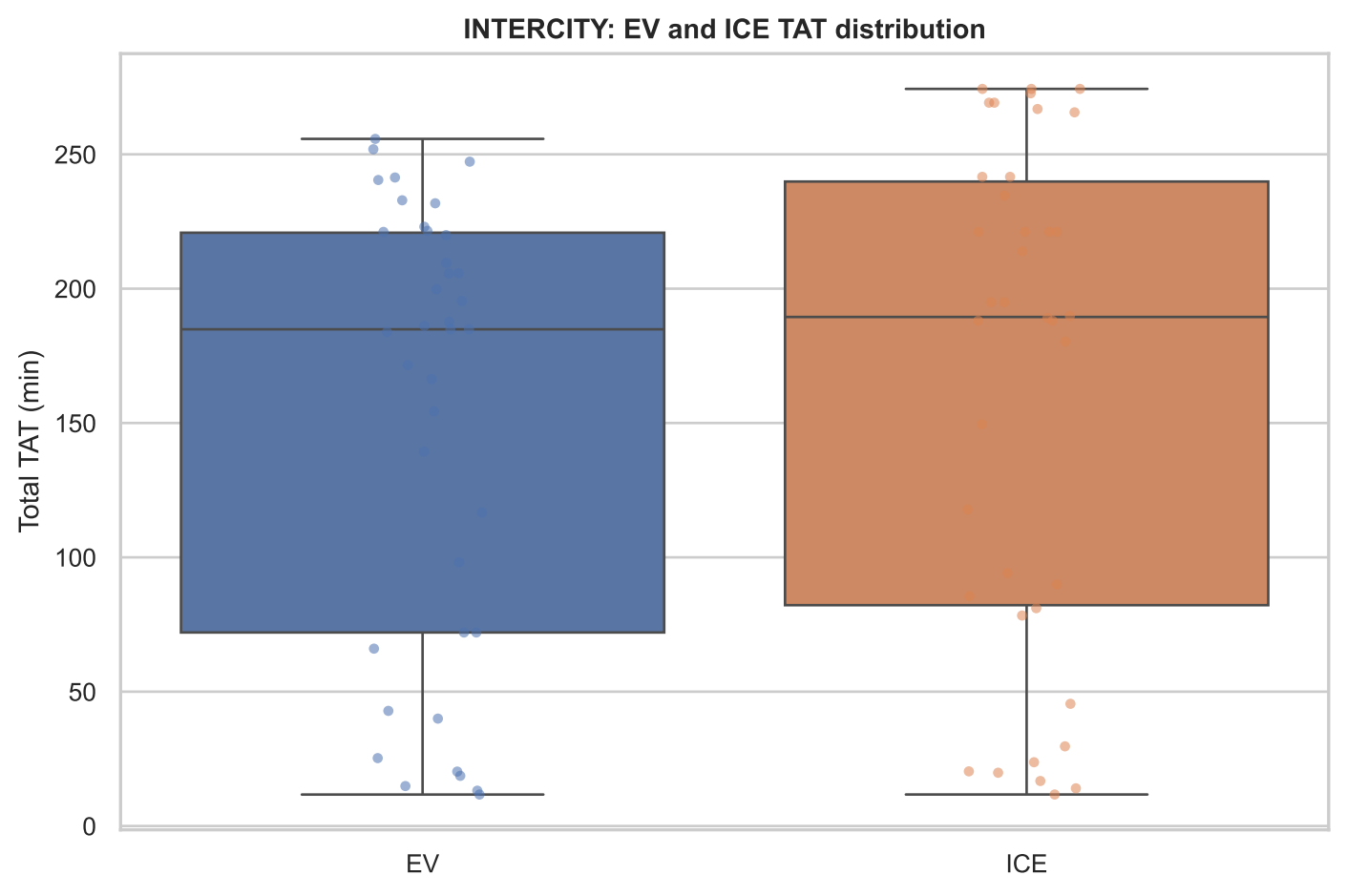
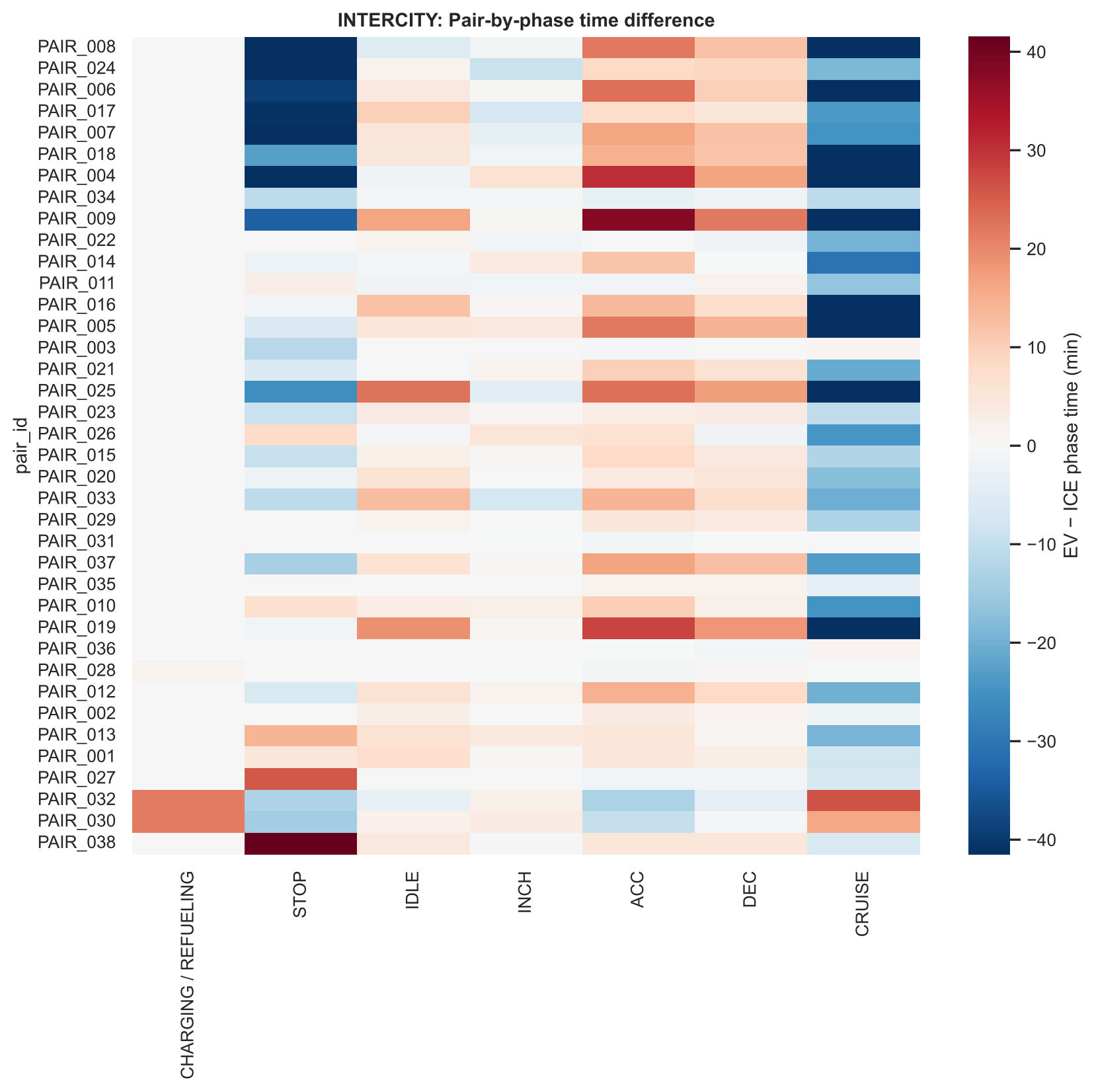
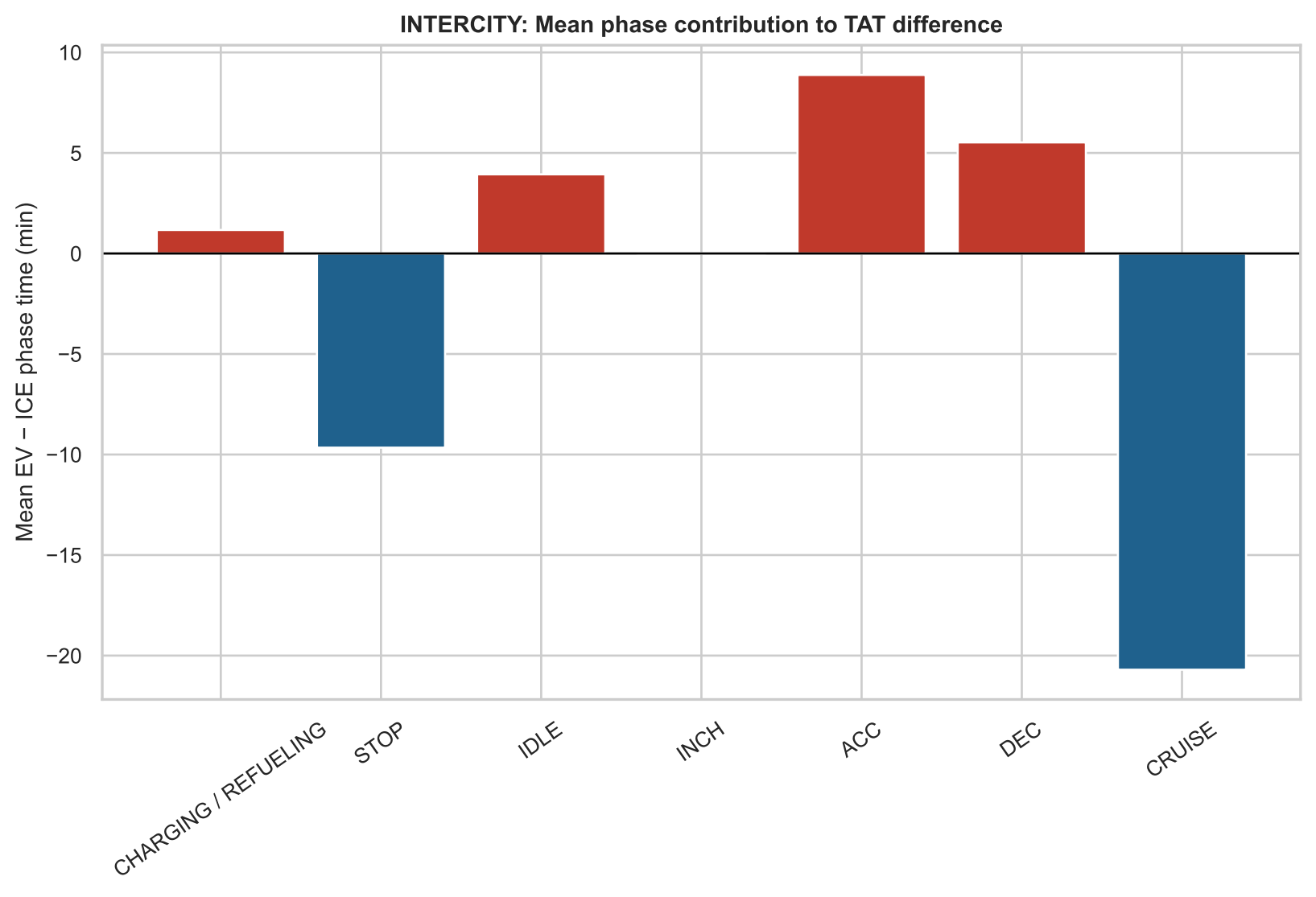
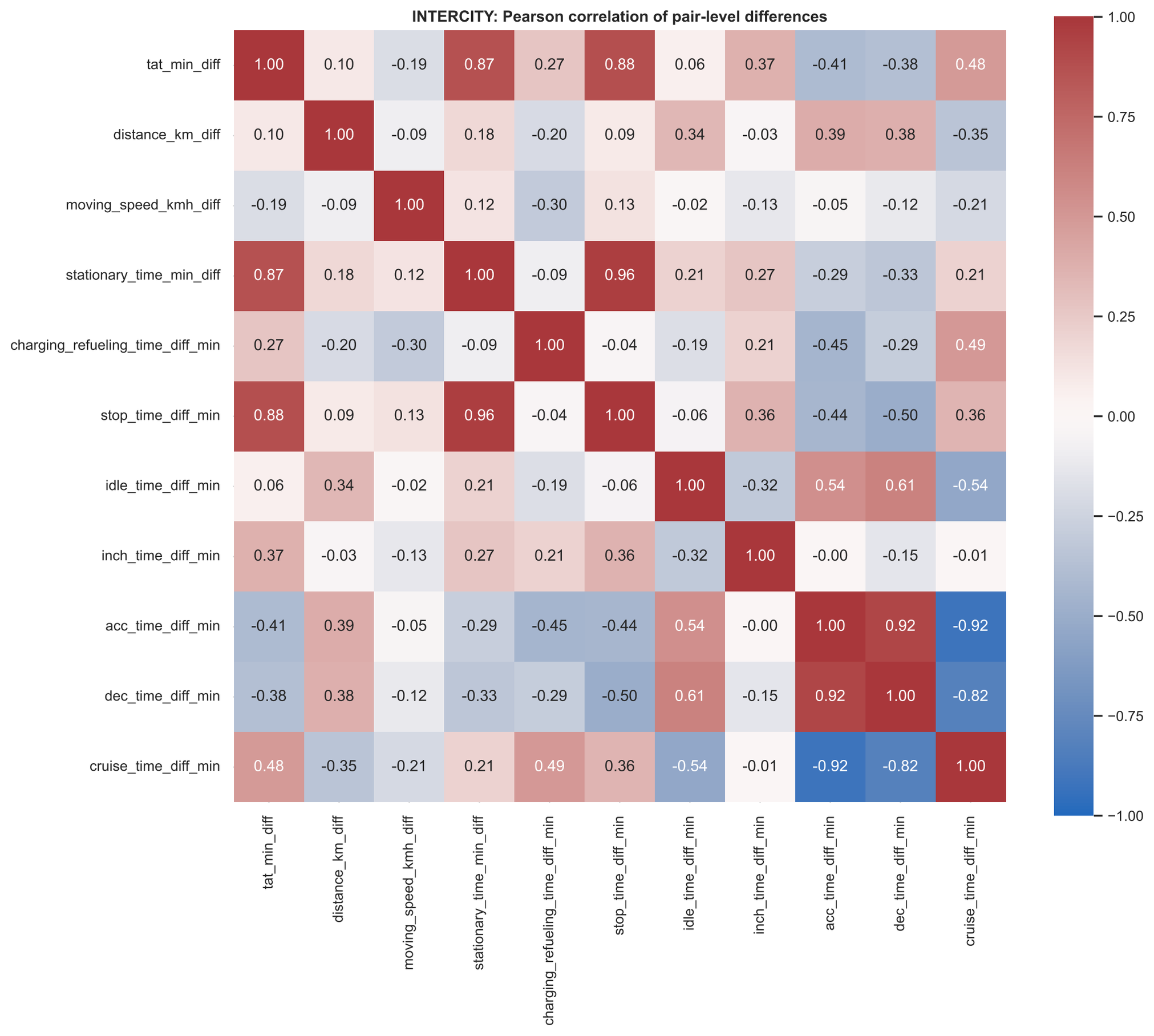

INTERCITY: Ranked pair-level TAT difference
PAIR_038
PAIR_030
PAIR_032
PAIR_027
PAIR_001
PAIR_013
PAIR_002
PAIR_012
PAIR_028
PAIR_036
PAIR_019
PAIR_010
PAIR_035
PAIR_037
PAIR_031
PAIR_029
PAIR_033
PAIR_020
PAIR_015
PAIR_026
PAIR_023
PAIR_025
PAIR_021
PAIR_003
PAIR_005
PAIR_016
PAIR_011
PAIR_014
PAIR_022
PAIR_009
PAIR_034
PAIR_004
PAIR_018
PAIR_007
PAIR_017
PAIR_006
PAIR_024
PAIR_008
−60
−40
−20
0
20
40
60
EV TAT − ICE TAT (min)

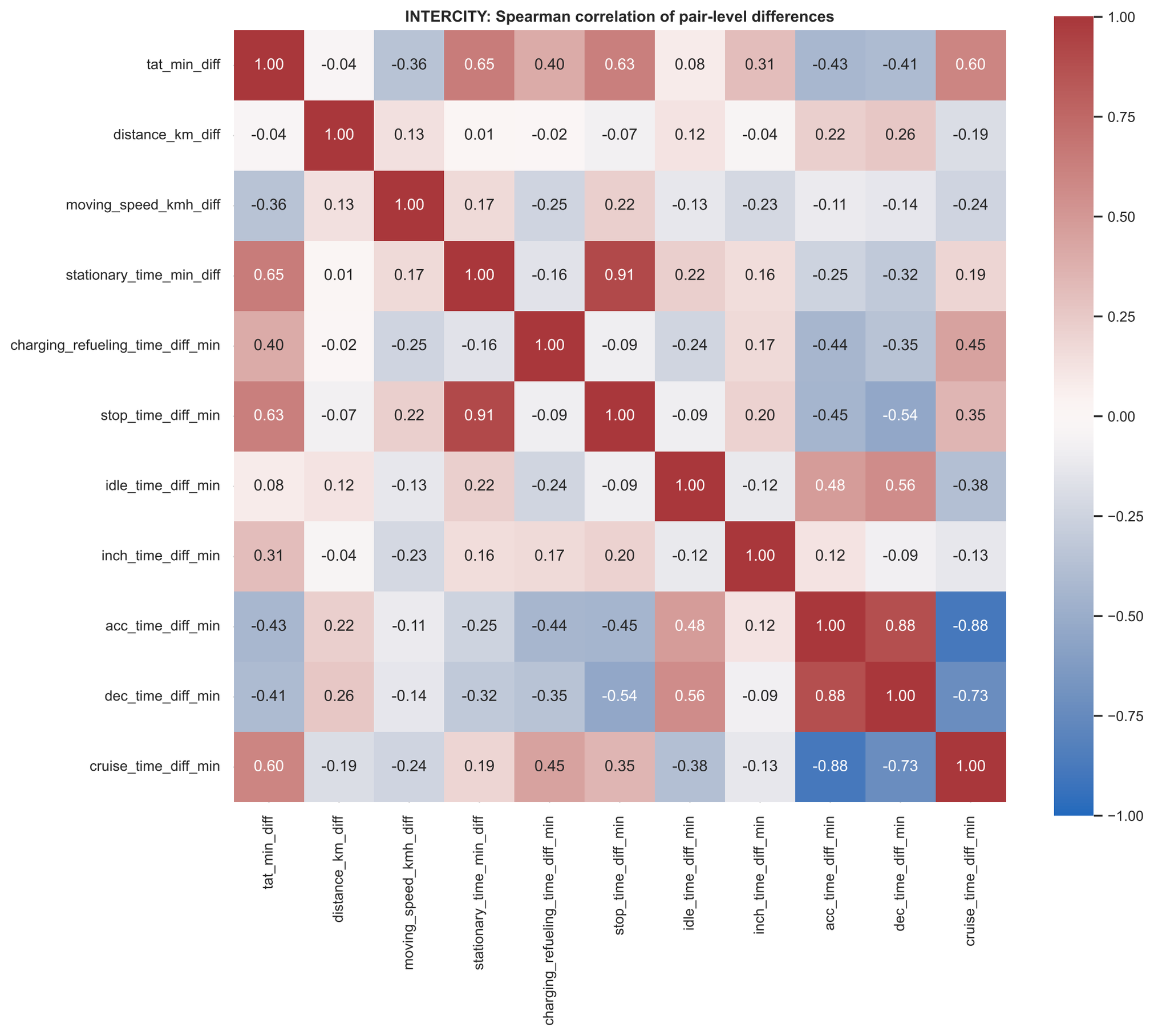

INTERCITY: Distribution of TAT differences
Count
0
2
4
6
8
10
−60
−40
−20
0
20
40
60
EV TAT − ICE TAT (min)

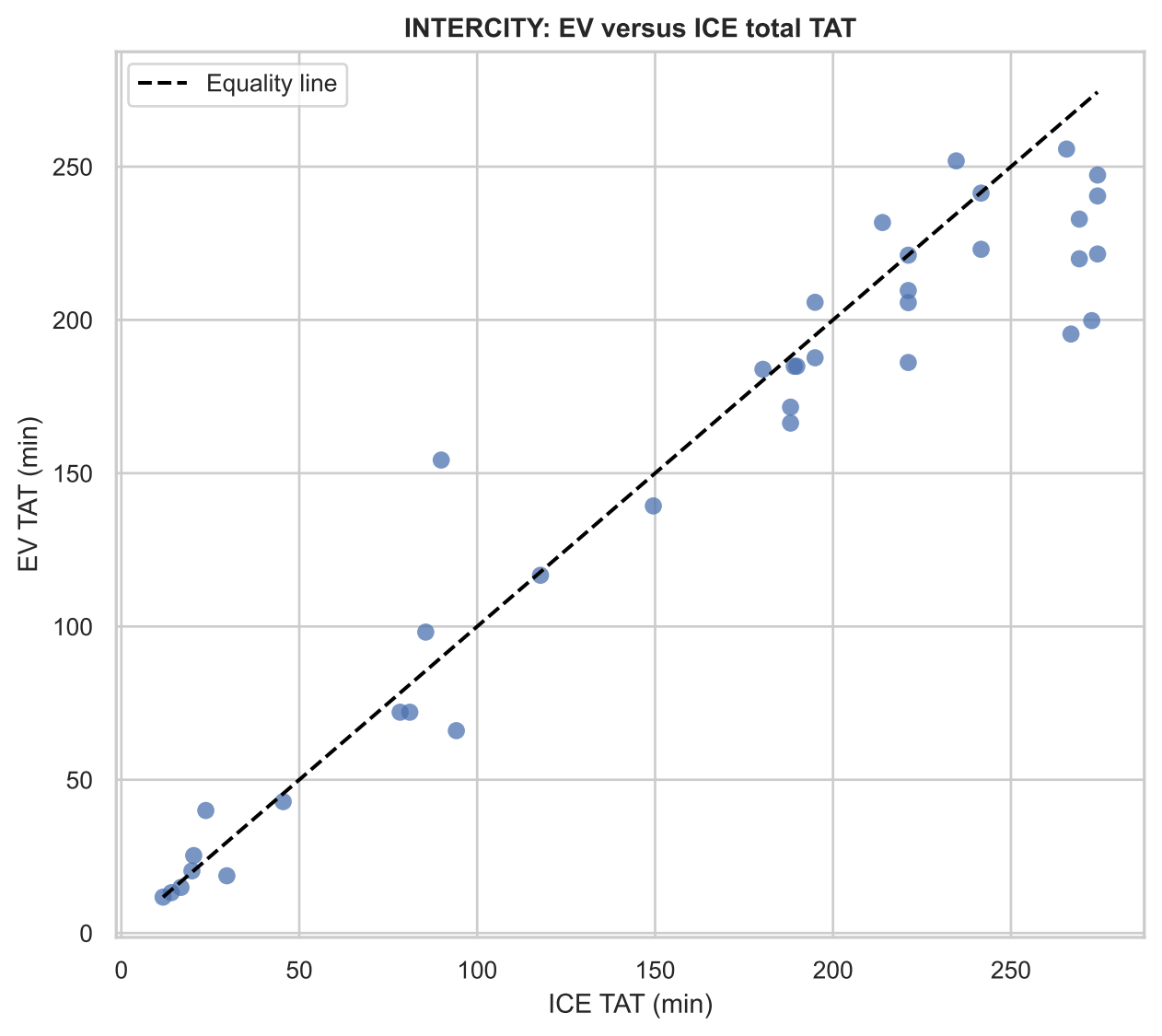

INTERCITY: EV and ICE TAT distribution
Total TAT (min)
0
50
100
150
200
250
EV
ICE

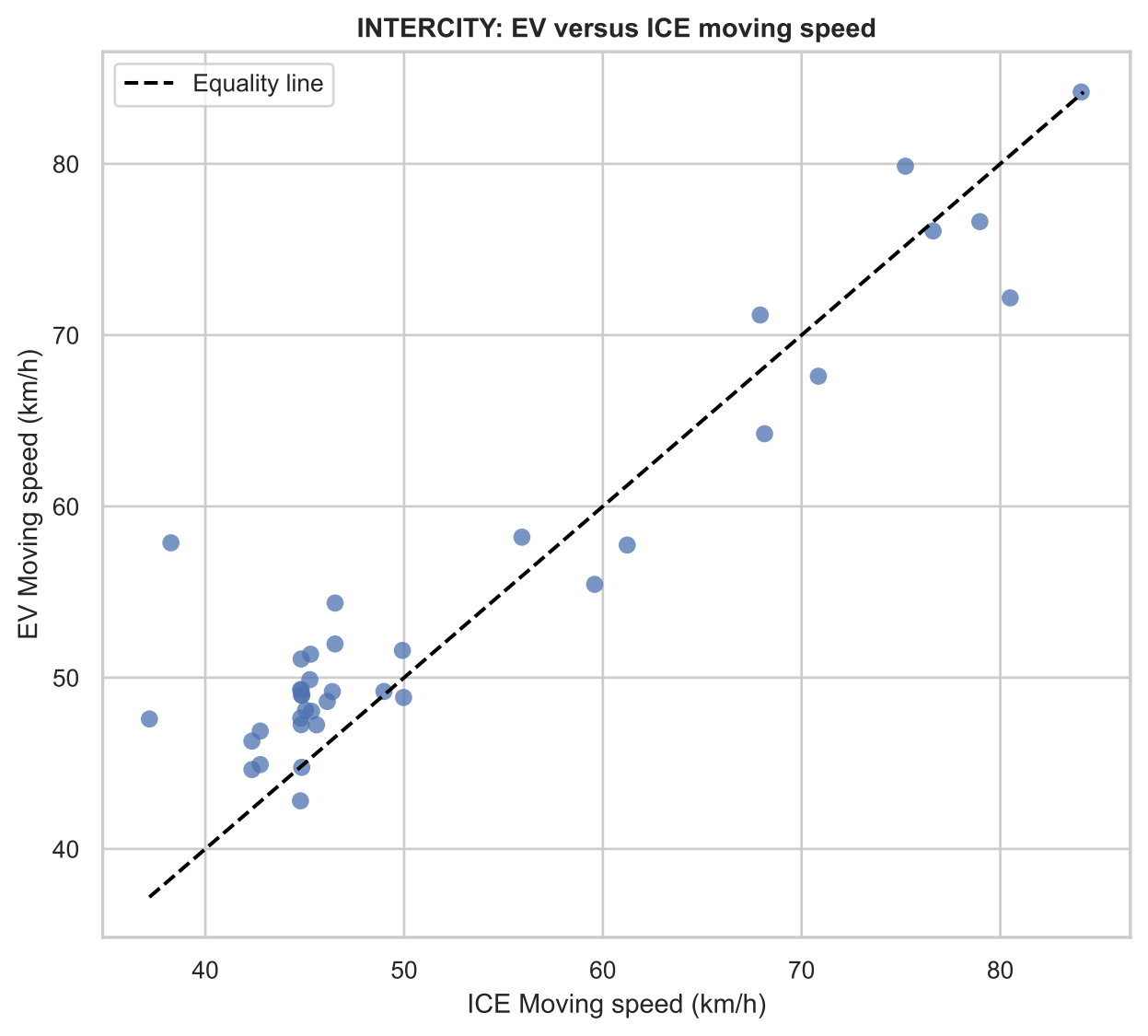

INTERCITY: Pair-by-phase time difference
pair_id
PAIR_008
PAIR_024
PAIR_006
PAIR_017
PAIR_007
PAIR_018
PAIR_004
PAIR_034
PAIR_009
PAIR_022
PAIR_014
PAIR_011
PAIR_016
PAIR_005
PAIR_003
PAIR_021
PAIR_025
PAIR_023
PAIR_026
PAIR_015
PAIR_020
PAIR_033
PAIR_029
PAIR_031
PAIR_037
PAIR_035
PAIR_010
PAIR_019
PAIR_036
PAIR_028
PAIR_012
PAIR_002
PAIR_013
PAIR_001
PAIR_027
PAIR_032
PAIR_030
PAIR_038
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
40
30
20
10
0
−10
−20
−30
−40

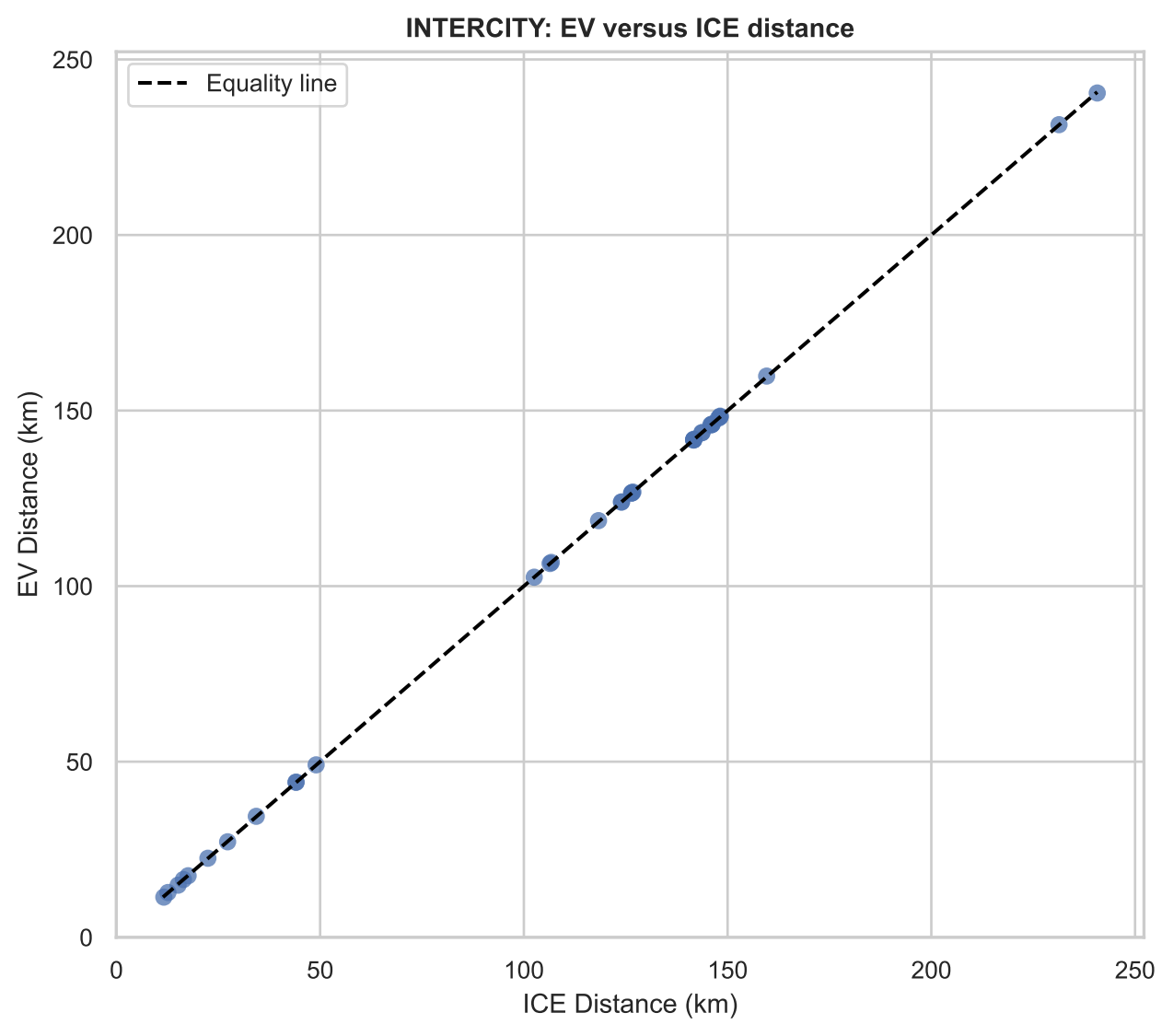

INTERCITY: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
10
5
0
−5
−10
−15
−20
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

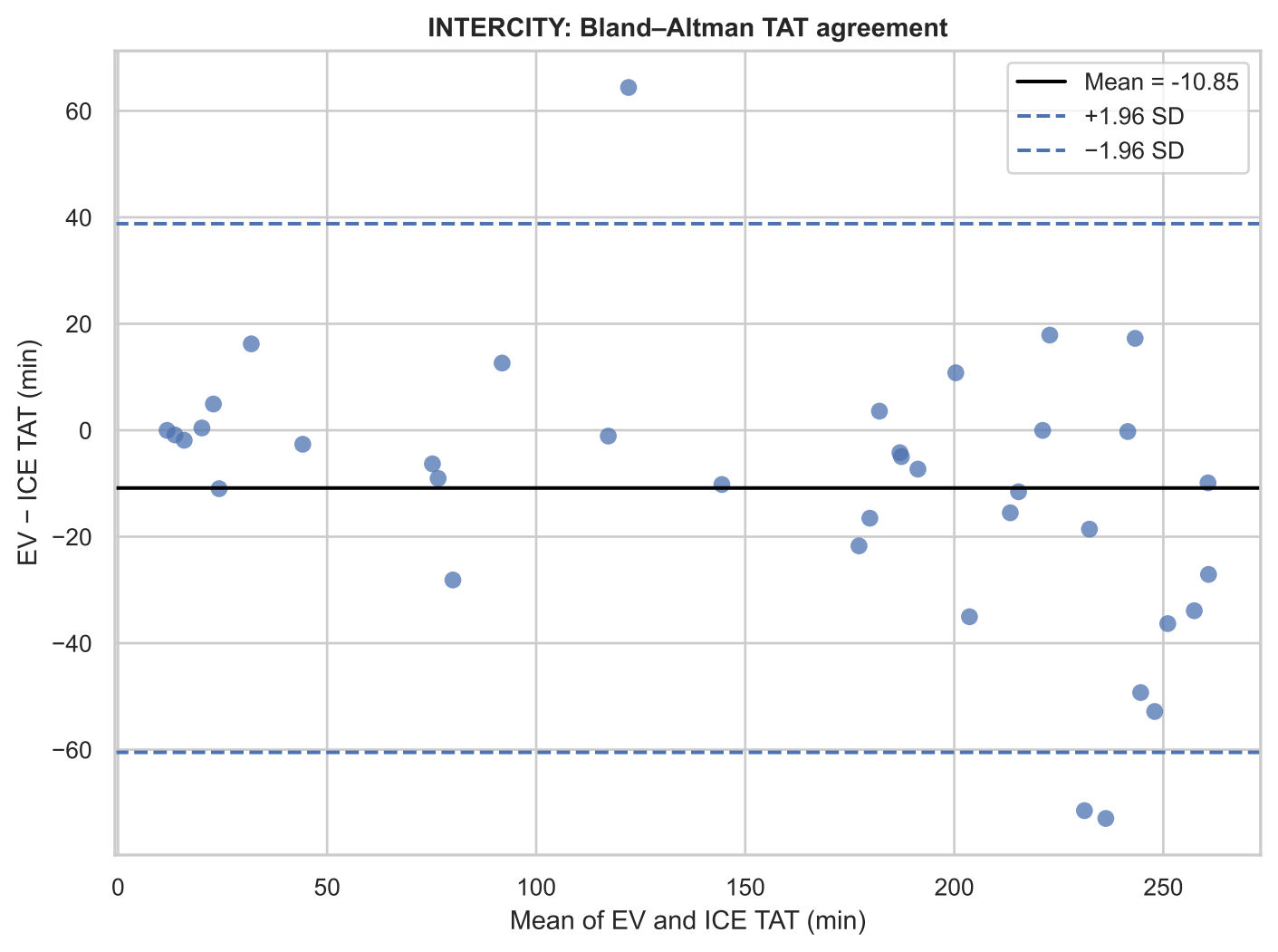

INTERCITY: Pearson correlation of pair-level differences
tat_min_diff 1.00 0.10 -0.19 0.87 0.27 0.88 0.06 0.37 -0.41 -0.38 0.48
distance_km_diff 0.10 1.00 -0.09 0.18 -0.20 0.09 0.34 -0.03 0.39 0.38 -0.35
moving_speed_kmh_diff -0.19 -0.09 1.00 0.12 -0.30 0.13 -0.02 -0.13 -0.05 -0.12 -0.21
stationary_time_min_diff 0.87 0.18 0.12 1.00 -0.09 0.96 0.21 0.27 -0.29 -0.33 0.21
charging_refueling_time_diff_min 0.27 -0.20 -0.30 -0.09 1.00 -0.04 -0.19 0.21 -0.45 -0.29 0.49
stop_time_diff_min 0.88 0.09 0.13 0.96 -0.04 1.00 -0.06 0.36 -0.44 -0.50 0.36
idle_time_diff_min 0.06 0.34 -0.02 0.21 -0.19 -0.06 1.00 -0.32 0.54 0.61 -0.54
inch_time_diff_min 0.37 -0.03 -0.13 0.27 0.21 0.36 -0.32 1.00 -0.00 -0.15 -0.01
acc_time_diff_min -0.41 0.39 -0.05 -0.29 -0.45 -0.44 0.54 -0.00 1.00 0.92 -0.92
dec_time_diff_min -0.38 0.38 -0.12 -0.33 -0.29 -0.50 0.61 -0.15 0.92 1.00 -0.82
cruise_time_diff_min 0.48 -0.35 -0.21 0.21 0.49 0.36 -0.54 -0.01 -0.92 -0.82 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

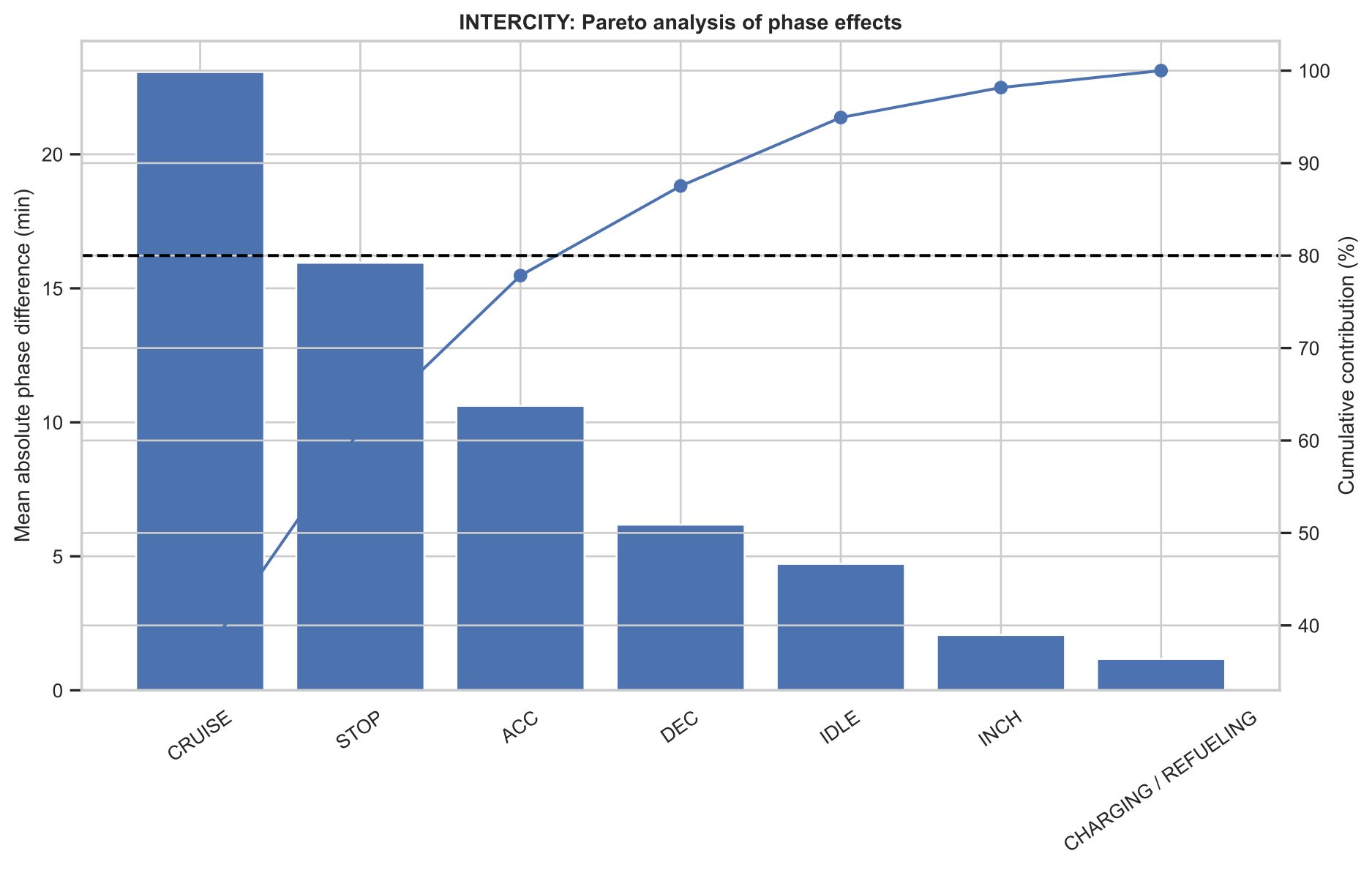

INTERCITY: Spearman correlation of pair-level differences
tat_min_diff 1.00 -0.04 -0.36 0.65 0.40 0.63 0.08 0.31 -0.43 -0.41 0.60
distance_km_diff -0.04 1.00 0.13 0.01 -0.02 -0.07 0.12 -0.04 0.22 0.26 -0.19
moving_speed_kmh_diff -0.36 0.13 1.00 0.17 -0.25 0.22 -0.13 -0.23 -0.11 -0.14 -0.24
stationary_time_min_diff 0.65 0.01 0.17 1.00 -0.16 0.91 0.22 0.16 -0.25 -0.32 0.19
charging_refueling_time_diff_min 0.40 -0.02 -0.25 -0.16 1.00 -0.09 -0.24 0.17 -0.44 -0.35 0.45
stop_time_diff_min 0.63 -0.07 0.22 0.91 -0.09 1.00 -0.09 0.20 -0.45 -0.54 0.35
idle_time_diff_min 0.08 0.12 -0.13 0.22 -0.24 -0.09 1.00 -0.12 0.48 0.56 -0.38
inch_time_diff_min 0.31 -0.04 -0.23 0.16 0.17 0.20 -0.12 1.00 0.12 -0.09 -0.13
acc_time_diff_min -0.43 0.22 -0.11 -0.25 -0.44 -0.45 0.48 0.12 1.00 0.88 -0.88
dec_time_diff_min -0.41 0.26 -0.14 -0.32 -0.35 -0.54 0.56 -0.09 0.88 1.00 -0.73
cruise_time_diff_min 0.60 -0.19 -0.24 0.19 0.45 0.35 -0.38 -0.13 -0.88 -0.73 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

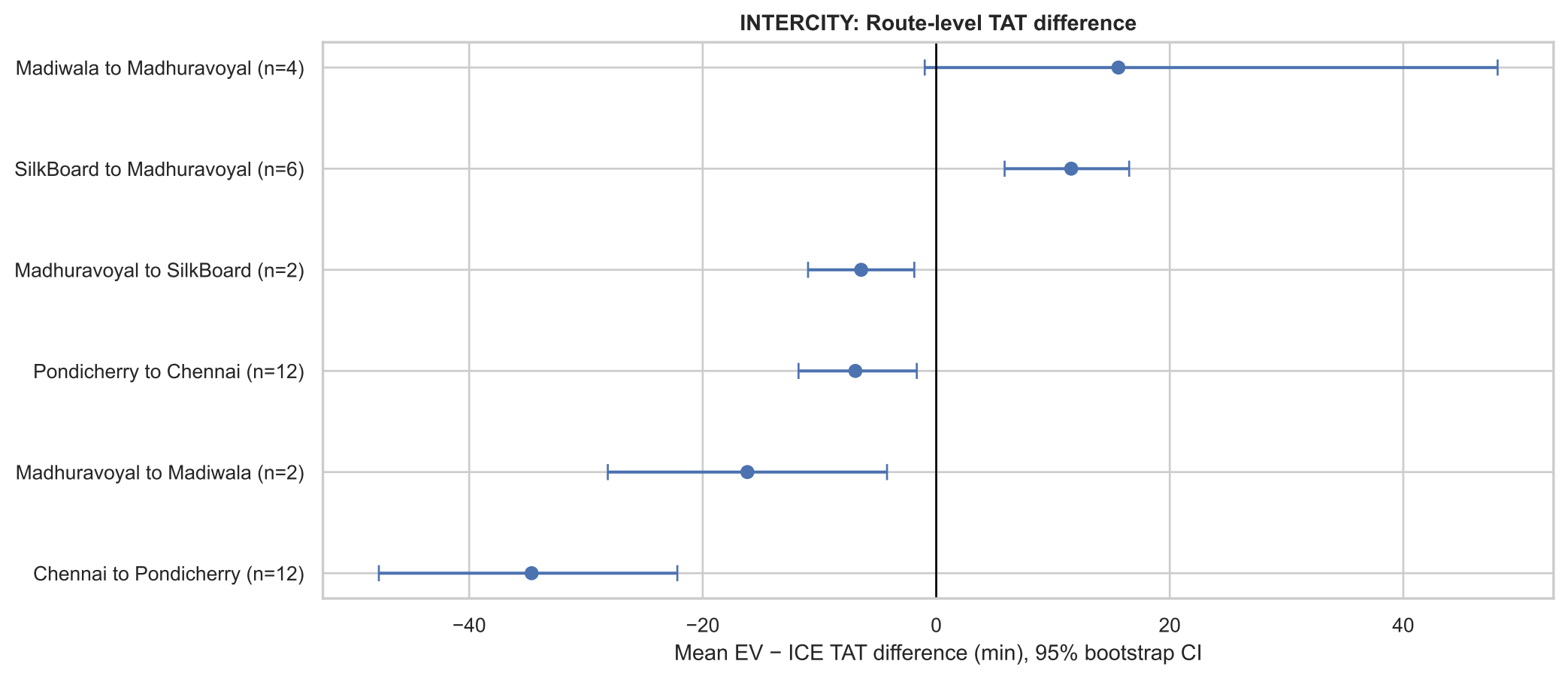

INTERCITY: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
0
50
100
150
200
250

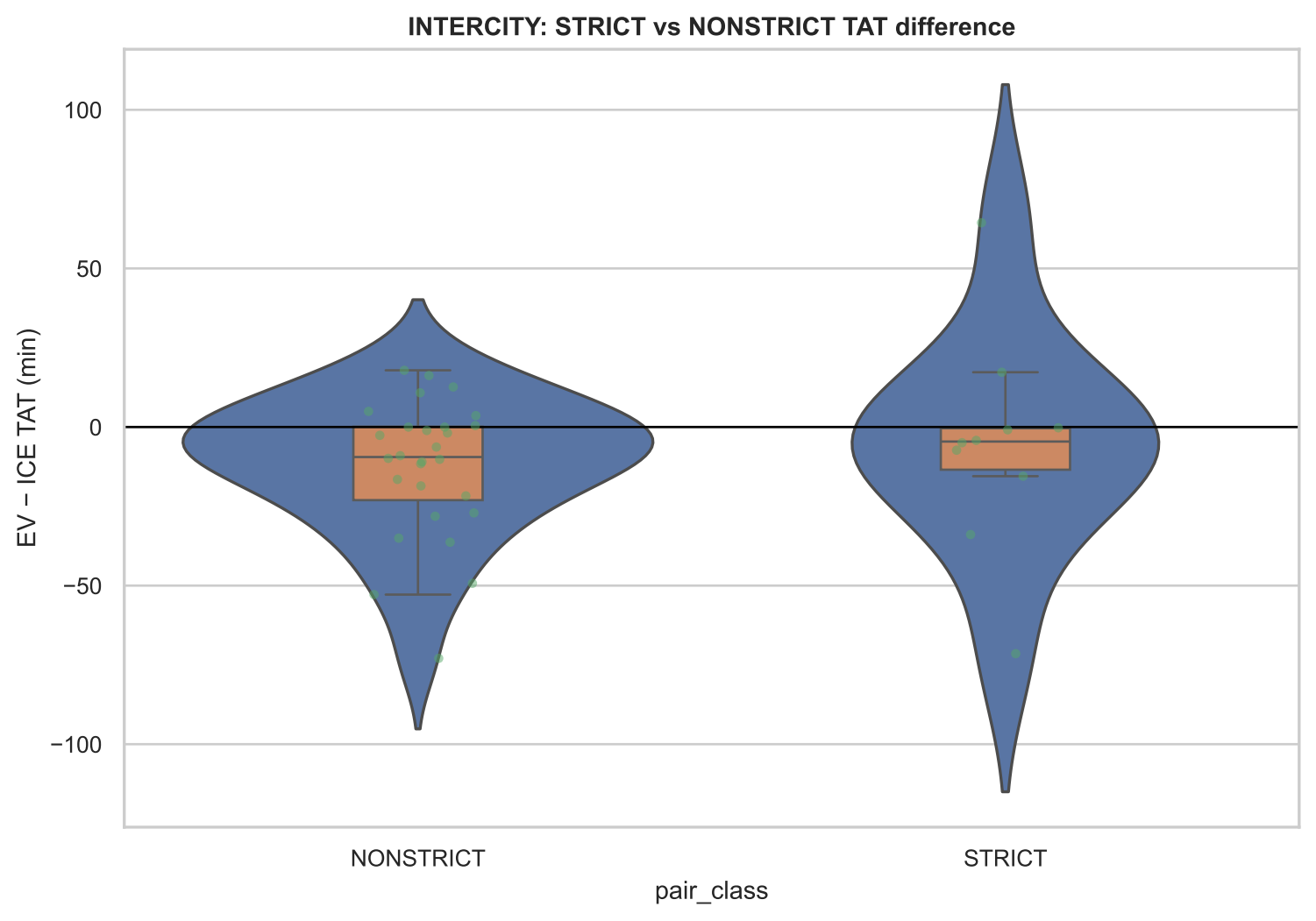

INTERCITY: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
40
50
60
70
80

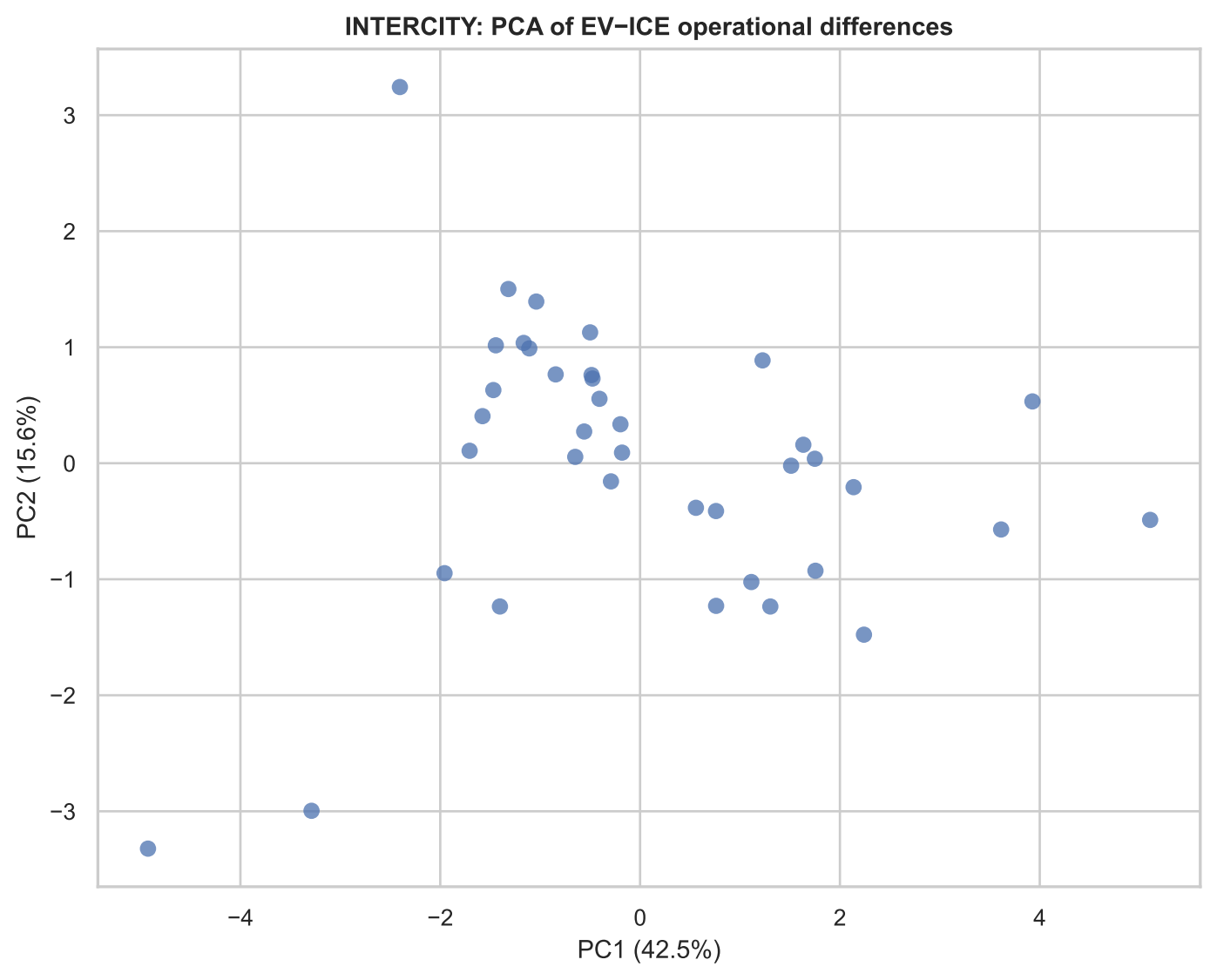

INTERCITY: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
0
50
100
150
200
250

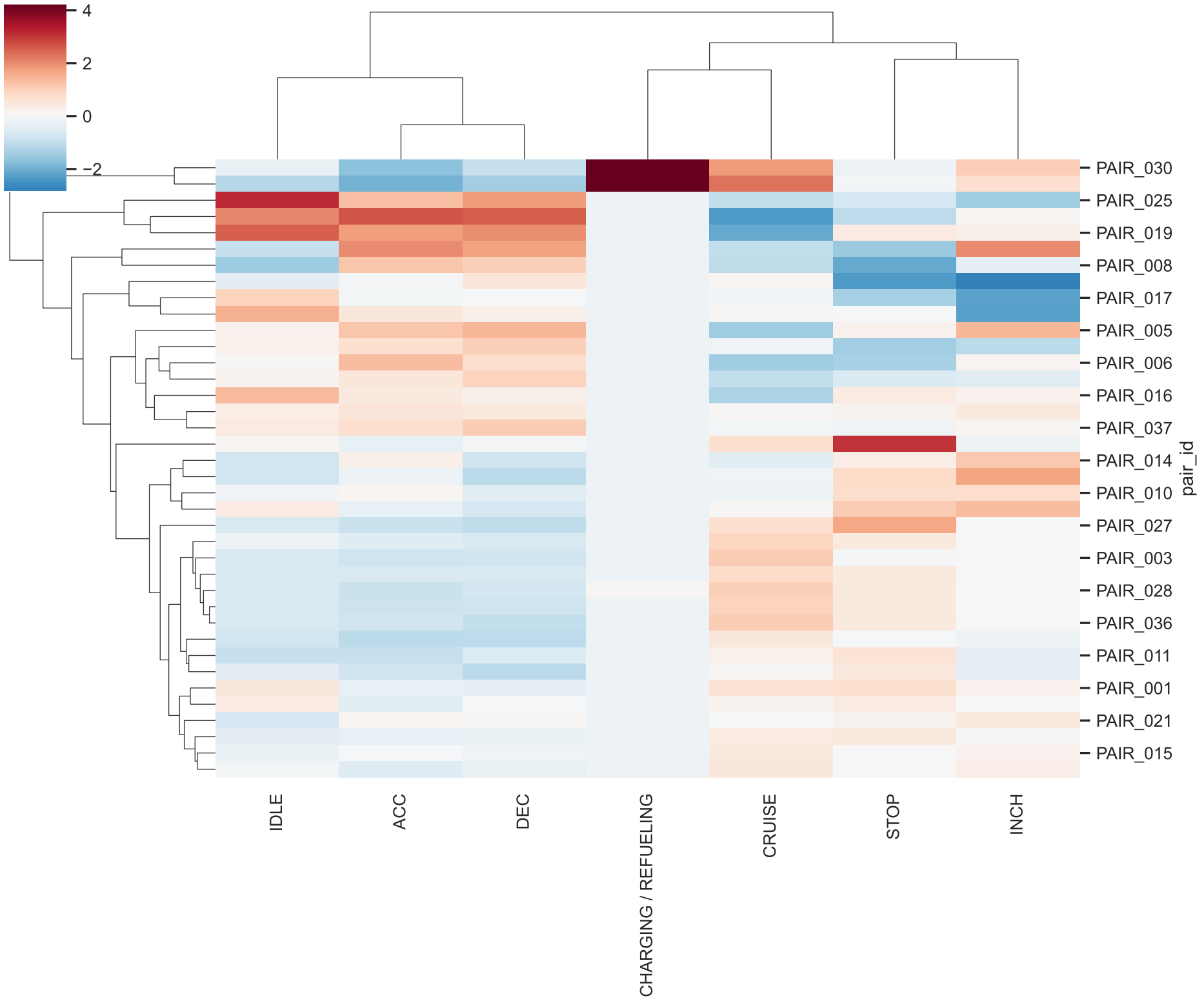

INTERCITY: Bland–Altman TAT agreement
Mean = -10.85
+1.96 SD
−1.96 SD
60
40
20
0
−20
−40
−60
EV – ICE TAT (min)
0
50
100
150
200
250
Mean of EV and ICE TAT (min)


INTERCITY: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
5
10
15
20
40
50
60
70
80
90
100
CRUISE
STOP
ACC
DEC
IDLE
INCH
CHARGING / REFUELING


INTERCITY: Route-level TAT difference
Madiwala to Madhuravoyal (n=4)
SilkBoard to Madhuravoyal (n=6)
Madhuravoyal to SilkBoard (n=2)
Pondicherry to Chennai (n=12)
Madhuravoyal to Madiwala (n=2)
Chennai to Pondicherry (n=12)
−40
−20
0
20
40
Mean EV − ICE TAT difference (min), 95% bootstrap CI


INTERCITY: STRICT vs NONSTRICT TAT difference
100
50
0
−50
−100
EV − ICE TAT (min)
NONSTRICT
STRICT
pair_class


INTERCITY: PCA of EV−ICE operational differences
PC2 (15.6%)
PC1 (42.5%)
3
2
1
0
−1
−2
−3
−4
−2
0
2
4


INTERCITY: Hierarchical clustering of phase-difference signatures
4
2
0
−2
PAIR_030
PAIR_025
PAIR_019
PAIR_008
PAIR_017
PAIR_005
PAIR_006
PAIR_016
PAIR_037
PAIR_014
PAIR_010
PAIR_027
PAIR_003
PAIR_028
PAIR_036
PAIR_011
PAIR_001
PAIR_021
PAIR_015
pair_id
IDLE
ACC
DEC
CHARGING / REFUELING
CRUISE
STOP
INCH